

Chapter 42: Matrix and linear transformations Worksheet

1. Are each of the following statements true or false?

(a) If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(\mathbf{x}) = A\mathbf{x}$ where A is a 3×4 matrix, then $n = 3$ and $m = 4$.

(b) If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a function with $T\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, then T cannot be a linear transformation.

(c) If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a function with $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, then T cannot be a linear transformation.

(d) $T : M_{nn} \rightarrow M_{nn}$ by $T(A) = A^{-1}$ is a linear transformation.

(e) If $T : V \rightarrow W$ is a linear transformation, and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V , then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is a basis for W .

2. Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation with $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ and $T\left(\begin{bmatrix} 3 \\ -2 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$. Find $T\left(\begin{bmatrix} -3 \\ 7 \end{bmatrix}\right)$. (Hint: Is $\begin{bmatrix} -3 \\ 7 \end{bmatrix}$ a linear combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$?)

3. Using the definition of linear transformation, determine whether the following function is a linear transformation. If yes, prove it. If no, provide an explicit counterexample showing that one of the linearity property fails.

If the function is a linear transformation, find the matrix A so that $L(\mathbf{x}) = A\mathbf{x}$.

(a) $L : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $L \left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right) = \begin{bmatrix} u_1 + u_2 \\ u_2 \\ u_1 - u_2 \end{bmatrix}$.

(b) $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} u_1 + 6 \\ u_2 \\ u_3 \end{bmatrix}$.

(c) $L : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $L \left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right) = \begin{bmatrix} u_2 \\ u_1 \\ u_1^2 u_2 \end{bmatrix}$.

4. Define $L : P_1 \rightarrow P_2$ by $L(p(x)) = xp(x) + p(0)$.

(a) Compute $L(3x + 4)$.

(b) Show L is a linear transformation.

5. Let $T : P_1 \rightarrow P_1$ be a linear transformation for which we know $T(x + 1) = 2x + 3$ and $T(x - 1) = 3x - 2$.

(a) Find $T(6x - 4)$.

(b) Find $T(ax + b)$.